Battery Management System

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BMS Is A Critical Component In An EV

By: **Express Web Desk** | New Delhi |
Updated: April 21, 2022 7:34:56 pm



Asserting that the Centre was committed to ensuring the safety of citizens, Gadkari said: "companies may take advance action to recall all defective batches of vehicles immediately". (Source: Twitter)

On Wednesday, an **80-year-old man died when the battery of an electric scooter exploded** while being charged at his home in Telangana's Nizamabad district. Soon after the Nizamabad incident made headlines, electric scooter manufacturer **Pure EV announced its decision to recall 2,000 vehicles** belonging to models ETrance Plus and EPluto 7G.

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Fuel Gauge

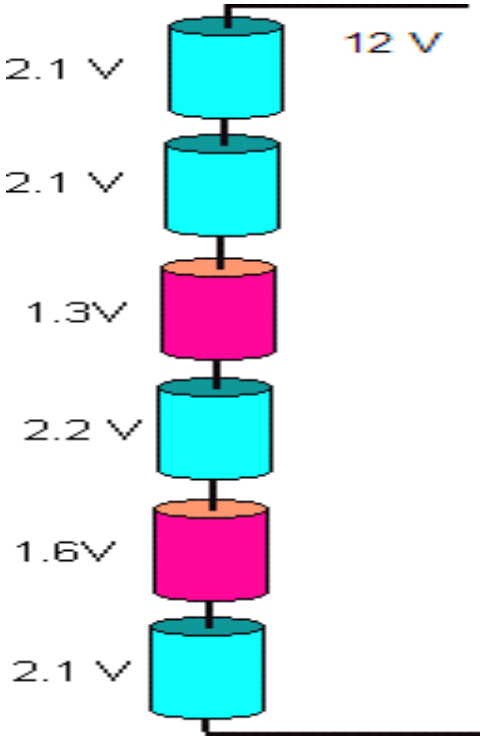
- Serves as cell/battery monitoring (no protection in-built)
- Dedicated IC based solution to measure voltage, temperature and current
- “fuel gauge” Integrated Circuit (IC) can work with simple methods to estimate the battery’s current State of Charge (SOC).

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Battery Protection

- Protecting a single cell is hard enough.

- Protecting a battery (a series string) is harder:
  - Cell voltages do not divide equally
  - Cell temperatures vary



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Analog BMS (“Protector”) PCM

- Voltage protection
- Current Protection
- Temperature Protection
- **May have:**
- Balancing
- **BUT**
- Doesn’t know what, where and by how much



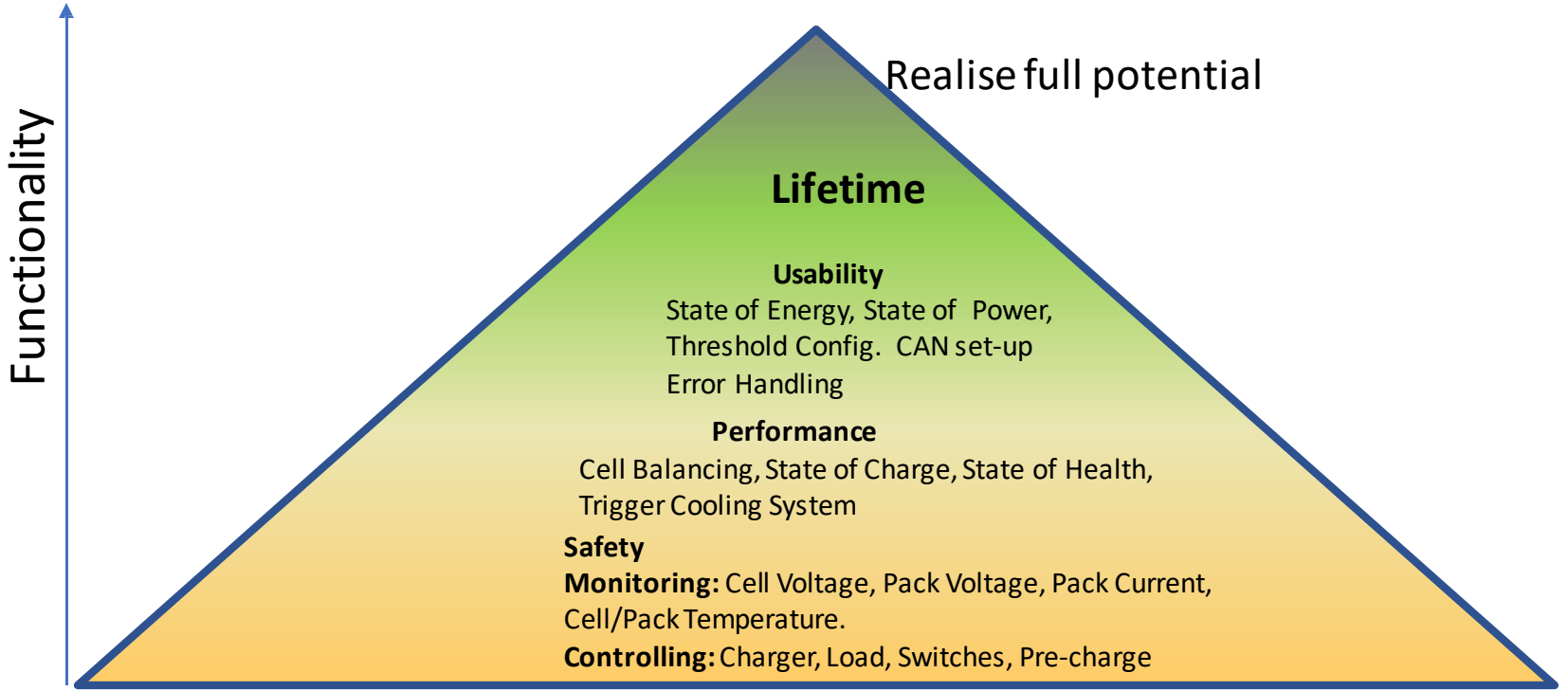
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Digital BMS

- Evaluation of State of Charge (“Fuel Gauge”)
- Evaluation of State Of Health
- Knows what, where and by how much
- Reports
- Requests shut down

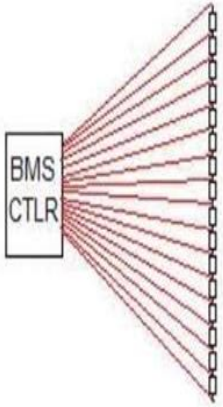
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BMS Functions



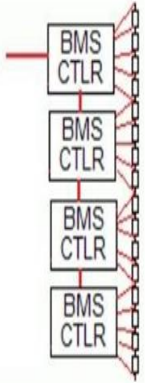
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BMS Topology – Application Based



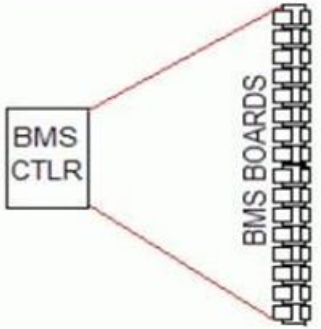
Centralized

A Single controller is connected to the cells multitude of wires



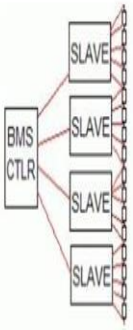
Modular

A few controllers, each handling certain number of cells with communication between the cells



Distributed

A BMS board is installed at each cells, with only communication cable between battery & controller



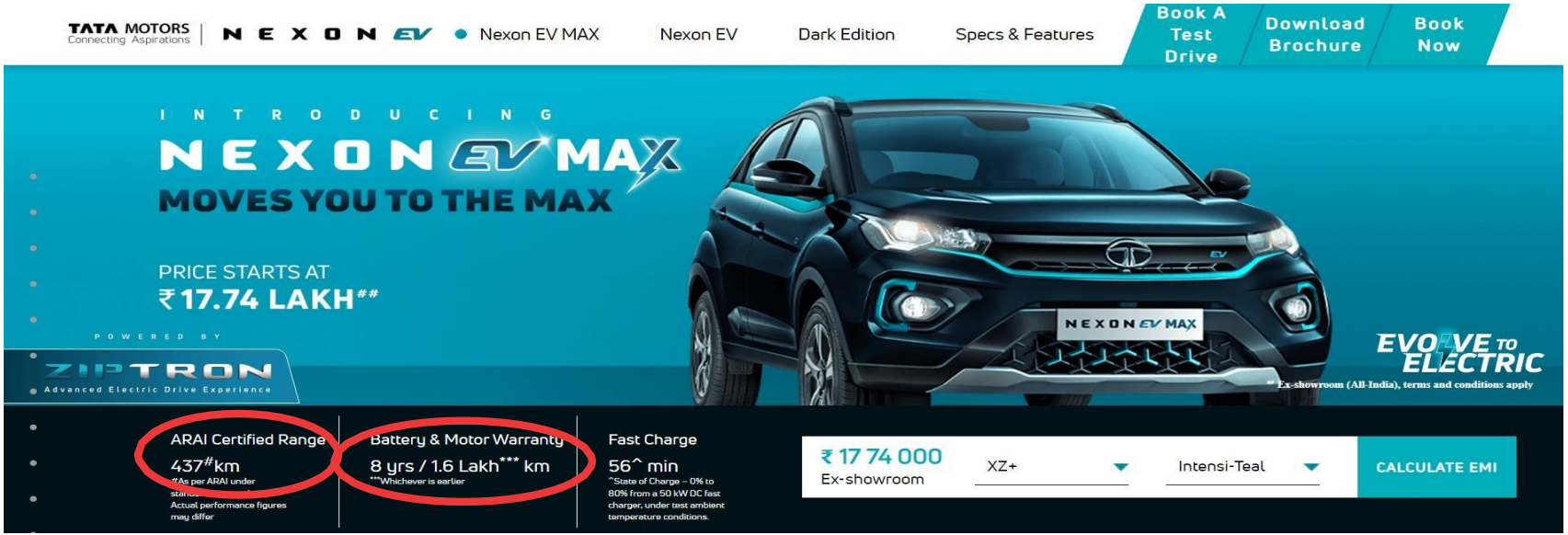
Master-slave

One master controller is connected to the slaves and the saves are connected to cells

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Good ‘Gatekeeper’

- The quality of the battery management system directly impacts the miles per charge an EV can deliver, maximizes the batteries overall lifetime, and, as a result, lowers the cost of ownership.



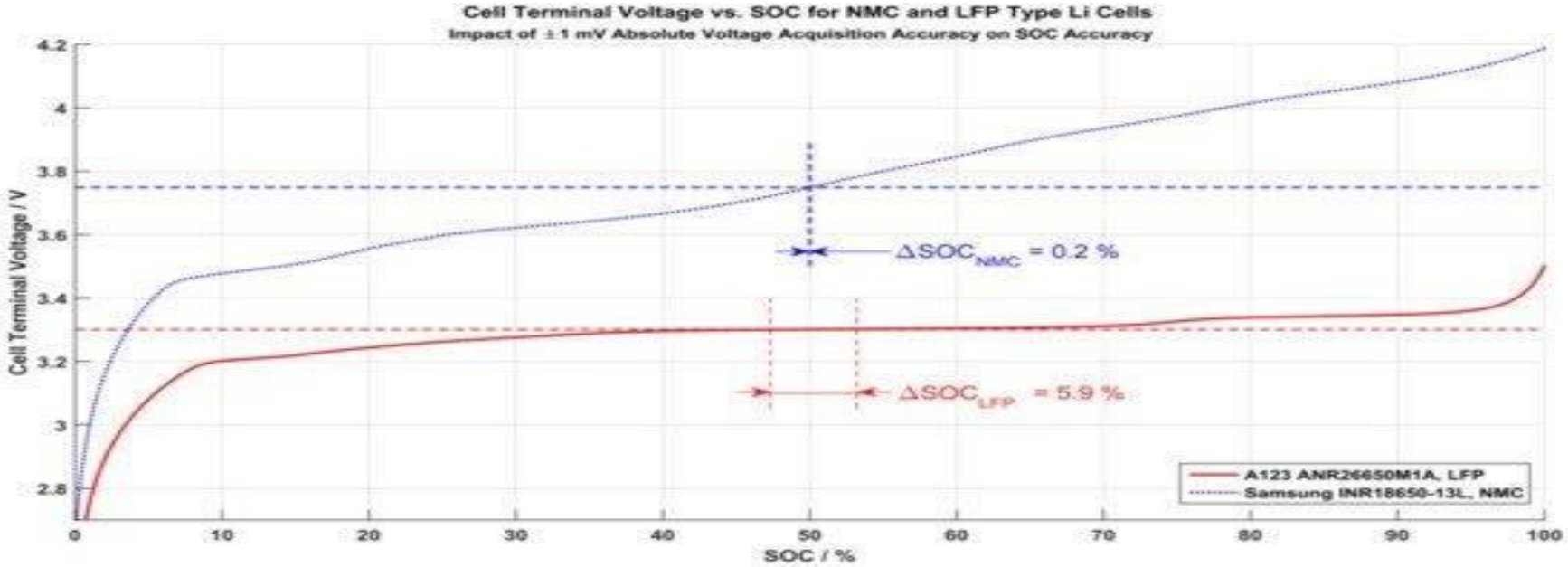
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Critical Functions

- Resolution of voltage and temperature measurement on cell level
- Accurate current sense on pack level
- System redundancy in order to allow implementation of the system
- on the highest safety level

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Impact of Voltage Measurement Accuracy



Source: https://www.mdpi.com/2076-3417/8/4/534/htm

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Application Specific IC (ASICs)

- Nearly all IC manufacturers have developed dedicated ICs that are fit for purpose for BMS application
- Solutions are available from vendors such as Linear Technologies (Analog
- Devices), NXP, ST Microelectronics
- **Measurement Accuracy:** 1.2mV Maximum Total Measurement Error, 16-Bit
- ADC (Analogue to Digital Converter) with Programmable Noise Filter
- **Measurement Speed:** 290µs to Measure All Cells in a System
- **SoC Calculation:** Synchronized Voltage and Current Measurement
- **Functions:** Passive Cell Balancing with Programmable Timer, 5 General Purpose Digital I/O or Analog Inputs

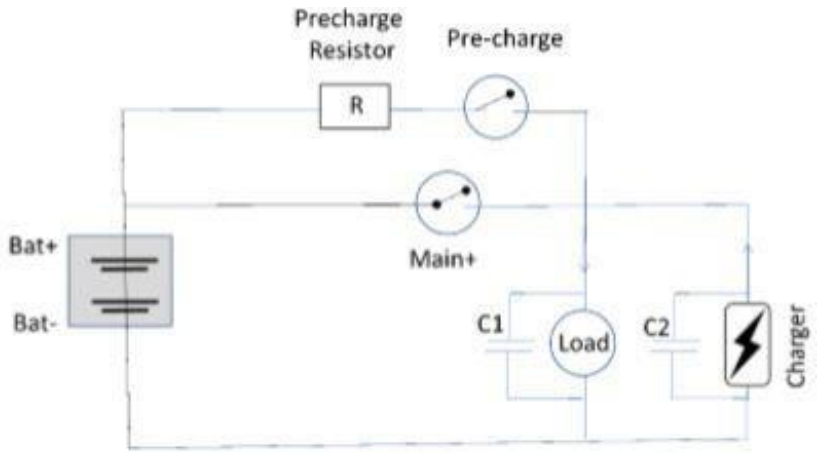
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Application Specific ICs

- **Environment:** Low EMI Susceptibility and Emissions, Uses a Single Twisted
  - Pair, up to 100 Meters
- **Usability:** Stackable Architecture for High Voltage Systems
- Built-in **isoSPI** (Isolated Serial Peripheral Interface) Communication Interface, 1Mb Isolated Serial Communications
- **Automotive Usage:** Engineered for **ISO 26262**-Compliant Systems, 4µA Sleep Mode Supply Current, **AEC-Q100** (Automotive Electronic Council)
  - Qualified for Automotive Applications…failure mechanism based stress test qualification for ICs

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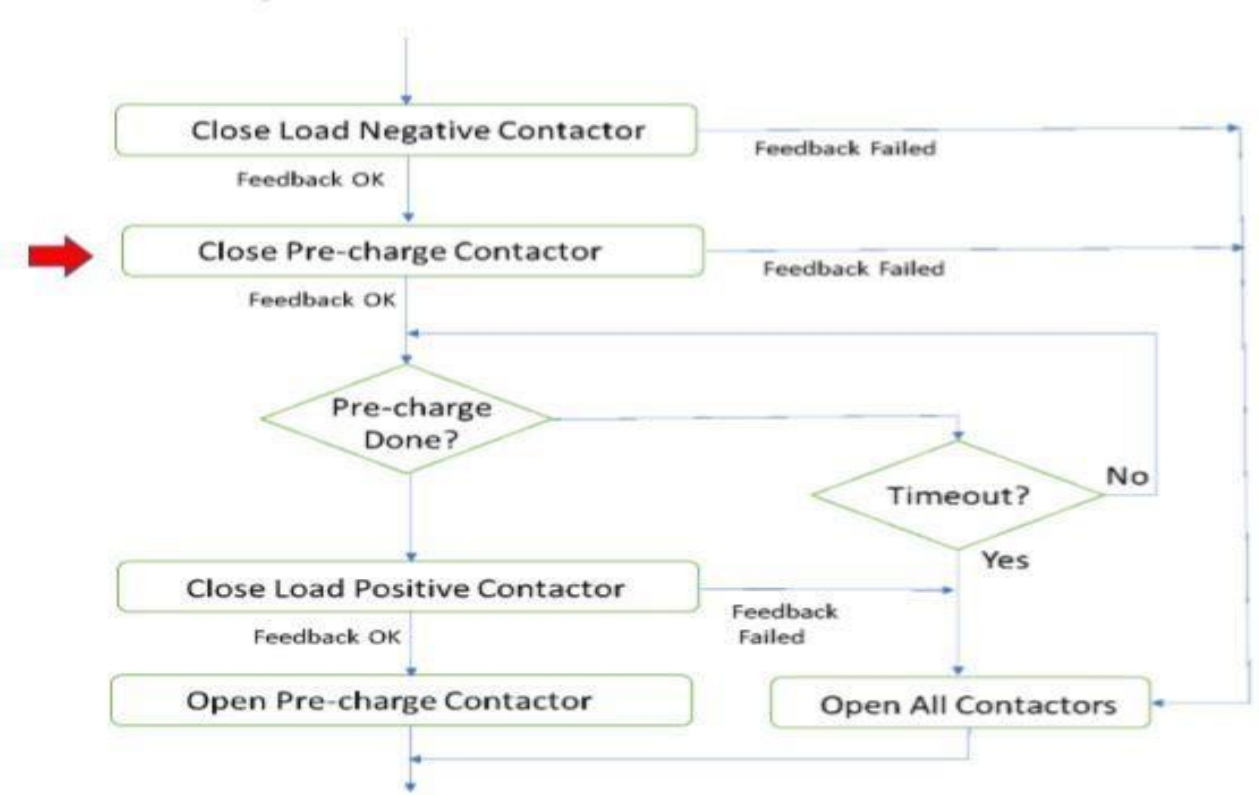
Pre-charge



- In most battery systems, the load and chargers have high capacitance at the high-current side facing the battery.
- These high capacitances (C1 and C2 on imply that the high in-rush currents maybe present for quite a while until the voltage in C1 or C2 matches the battery pack voltage.
- High current levels for extended periods of time can damage the system for example, the connection pads of the main contact or may weld.

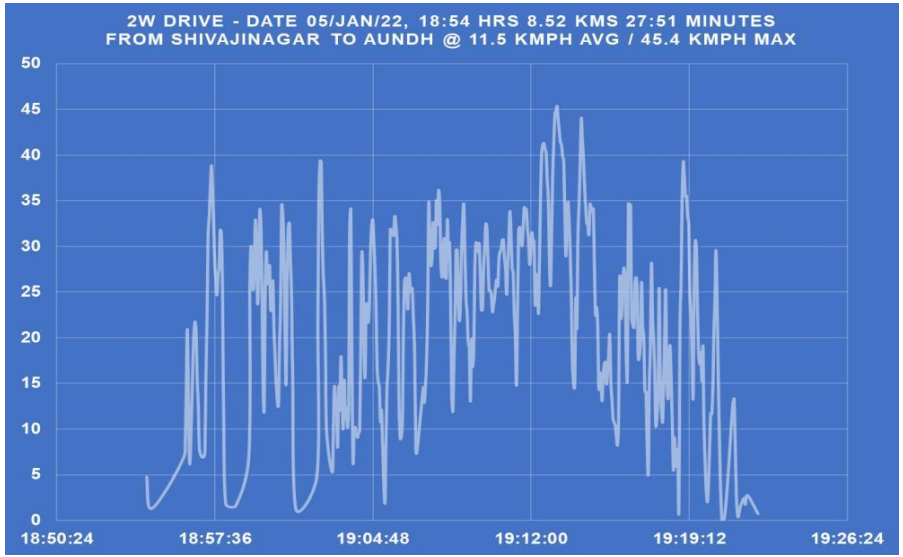
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Activating Load Sequence



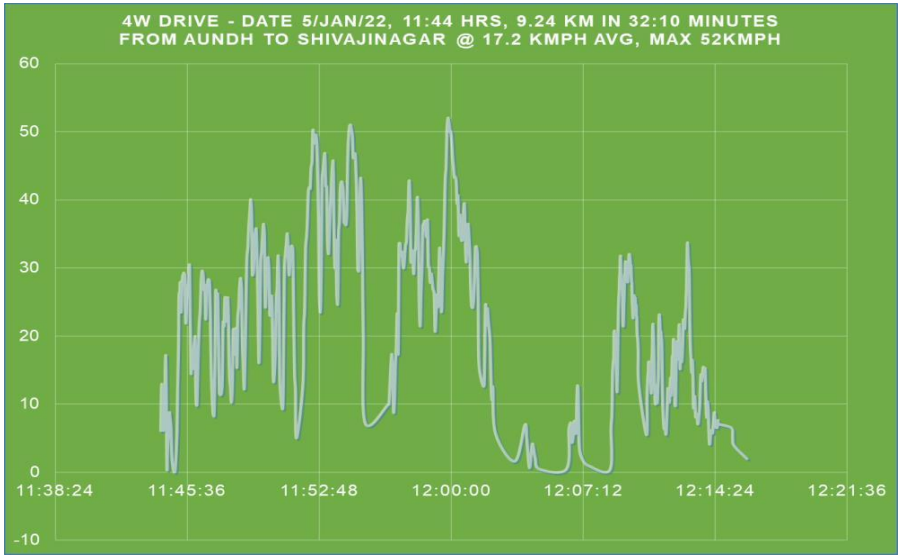
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Real Life Drive Cycle



2W Drive Cycle @ Pune

4W Drive Cycle @ Pune



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Modes of Operation

- **Idle mode:** BMS Performs all the functions except allowing current
- **Charge mode:** BMS performs all the functions and allows current from
  - Charger to the battery Pack
- **Discharge mode:** BMS performs all the functions and allows current from the battery Pack to the Load
- **Error Mode:** BMS Performs all the functions except allowing current and doesn't allow entering into Charge or Discharge mode unless all the Errors in the system are resolved

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What causes Imbalance

- Cells are unbalanced (unequal energy) because of:
  - **Initial Conditions**
    - Manufacturing and environment variance
  - **During Operation**
    - Imbalance between charging and discharging
    - Differences in thermal conditions
  - **Over Life**
    - Degradation with aging
    - Differences in internal impedance
    - Unequal self-discharging

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Cell Balancing

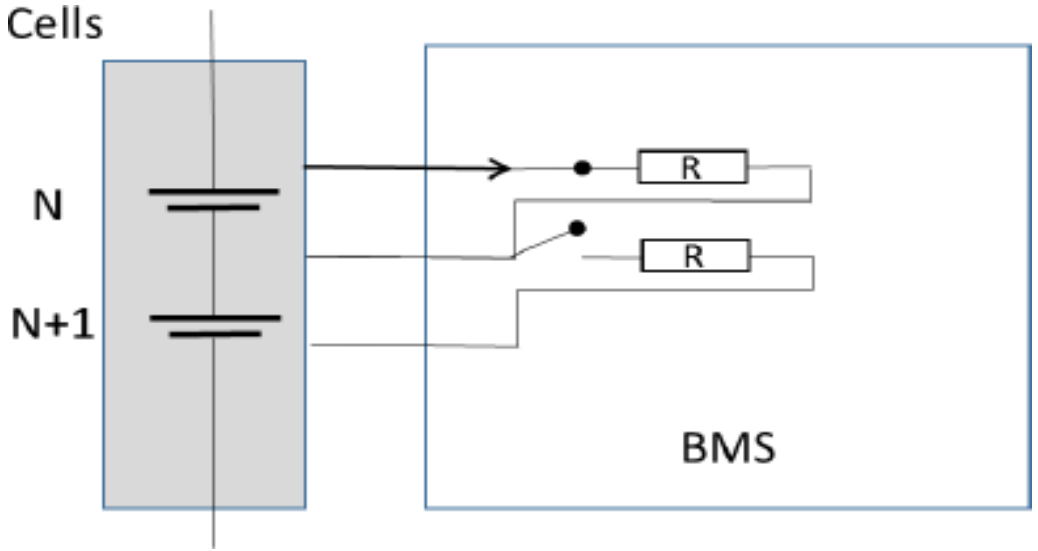
- Any Battery Pack is only as good as the weakest cell
- All cells equally charged = maximum available energy
- Balancing removes charge from fullest cells, to leave room for more charging,
    - so the other cells can catch-up



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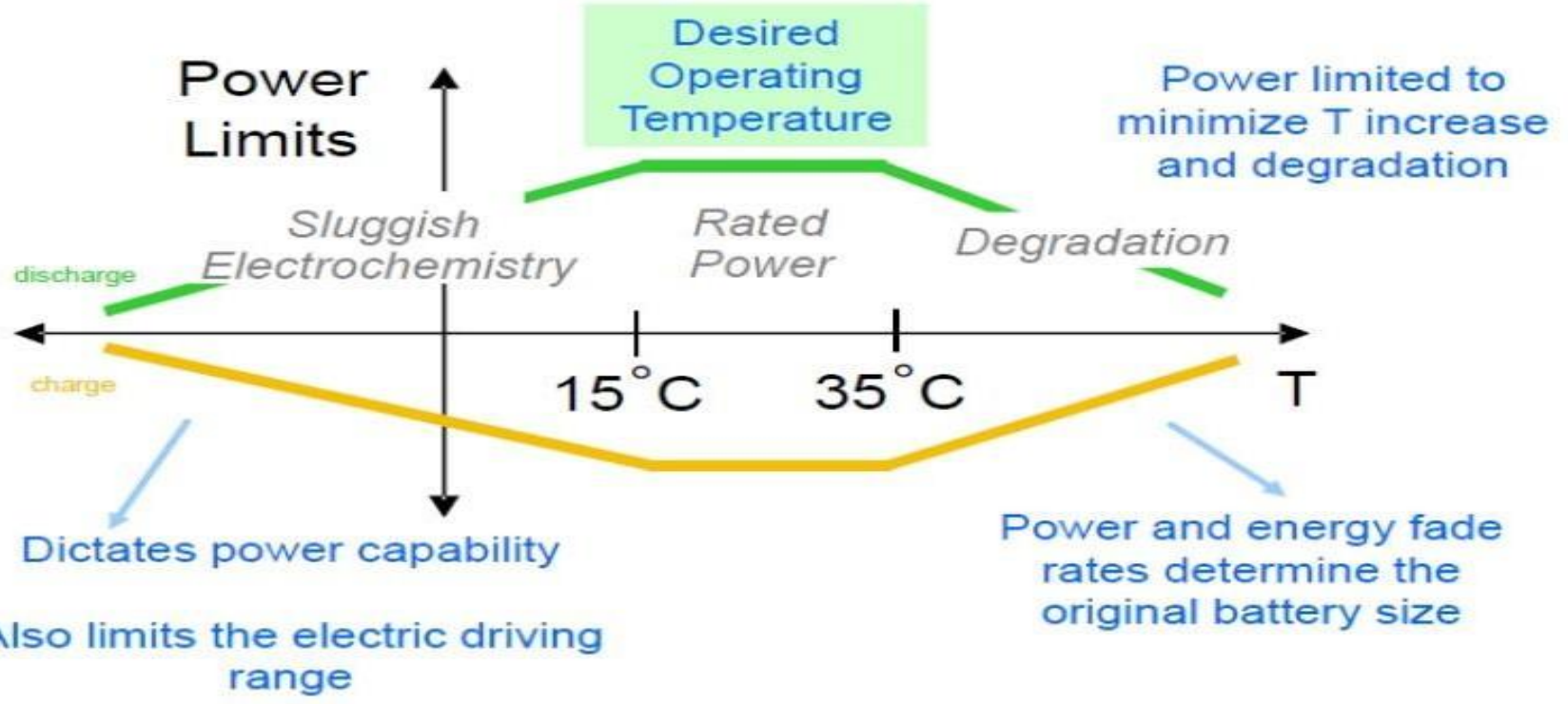
Cell Balancing

- Cell balancing is a key BMS function which ensures that when a battery is charged, all cells will be charged to their maximum capacity(100% SoC).
- Types of Cell Balancing:
  - Passive Balancing
  - Active Balancing



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Thermal Management

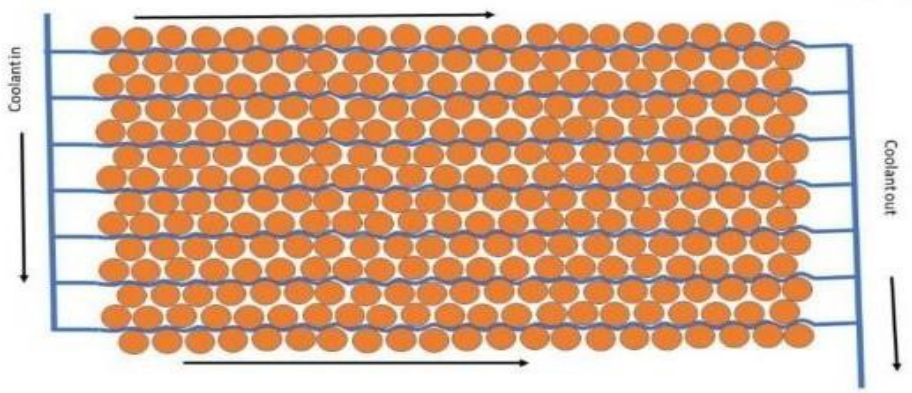


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Tesla Model 3

- Tesla model 3 features cell cooling. Individual cooling channels are sandwiched in between two cell layers
- The cooling channel covers whole height of the cell and 164 cells in a row
- Seven such rows are connected in parallel
- resulting in better cooling
- Coolant is also supplied to the power electronics and to the motor at the rear using a separate pump



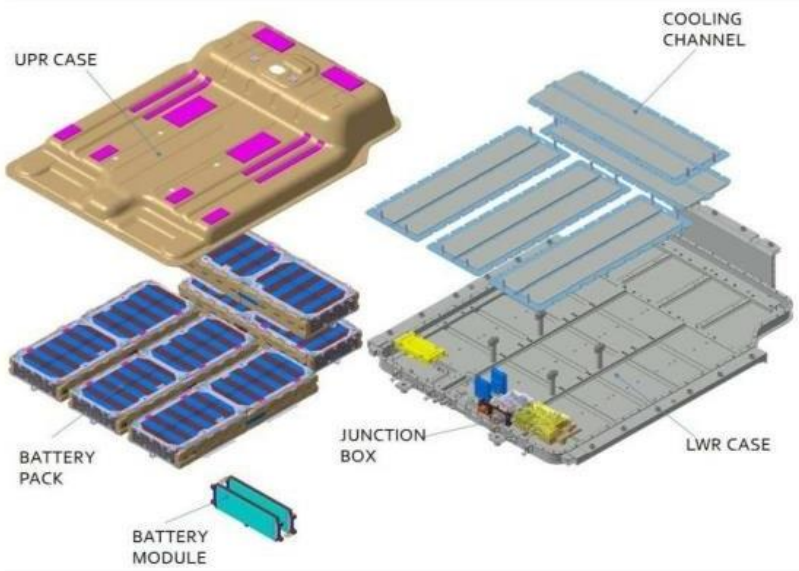


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Kona Electric

- Each module has a cooling plate attached at
- the bottom of the module
- The coolant used is water and ethylene glycol
- mix.
- It also houses a chiller mode for extremely hot conditions.
- For colder areas a heating can also be provided using a dedicated heater in temp range of 0 to 32 0F.
- The battery heater, if present, is located outside the battery pack and warms the liquid “coolant” just before it enters the battery pack.

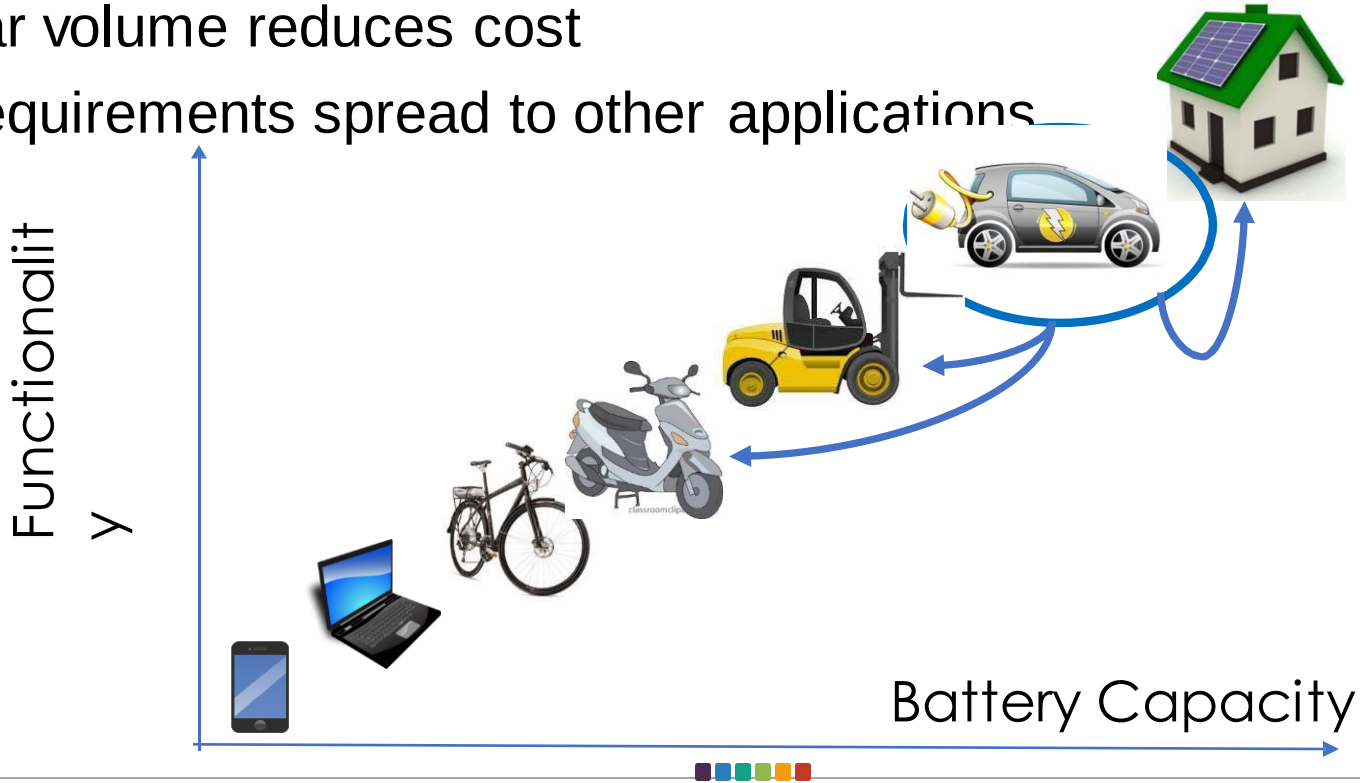




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BMS applications

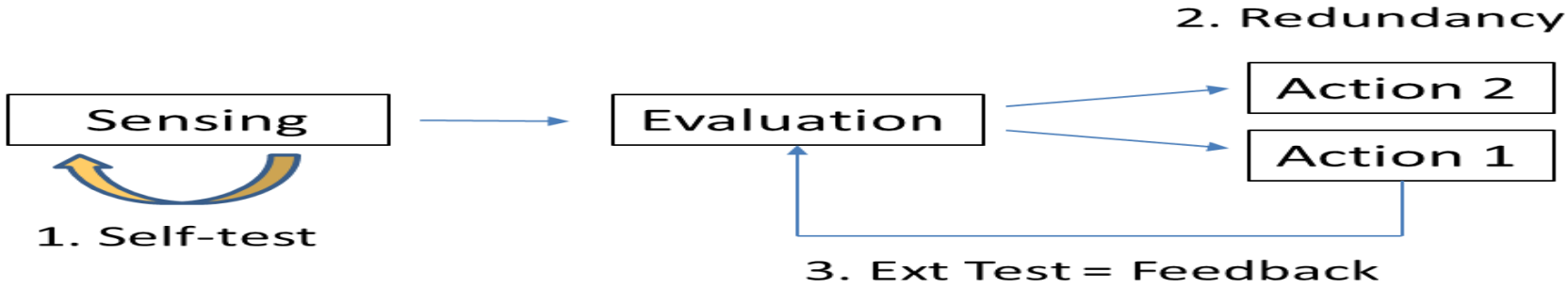
1. Cars drives BMS requirements
2. Car volume reduces cost
3. Requirements spread to other applications



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Functional safety

- ISO26262 ASIL (Automotive Safety Integrity Level) C the norm in automotive for BMS
- Spreading to industrial machines, 2 wheelers, energy storage
- **ASIL C** level requires BMS components at FIT rate below **1 FIT** (number of Failures In Time interval of 109 hours of operation
  - **MTBF of 1 billion hours**
- Very difficult to achieve, consequently "Safety Mechanisms" like "Testing" and "Redundancy" are used:



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Functional Safety - Example

- No safety critical undetected singlepoint failures

- This is achieved by:
  - Cell voltage input open wire detection
  - Cell voltage and temperature measurement self test
  - Internal and external MUX self test
  - Power supply with watchdog protection, self test and CRC
  - **Dual-core Lockstep cycle-by-cycle CPU**
  - CPU built in self test
  - ECC on CPU FLASH and RAM
  - CPU Clock and Voltage monitoring
  - Pre-qualified safety rated key components
  - Pre-qualified RTOS (Real Time Operating System)

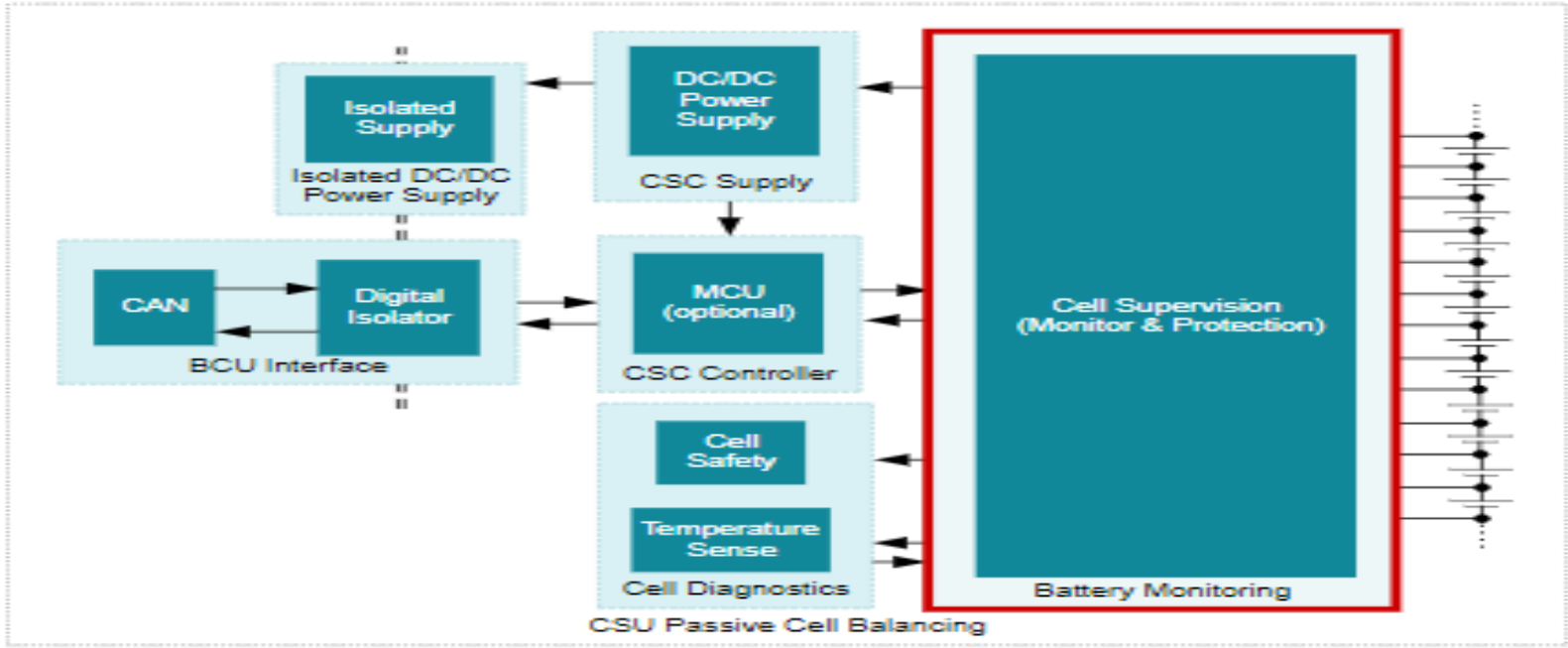
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Advanced models for more Accurate SOC

1. Coulomb counting
2. Combining OCV and CC
3. Kalman filter
4. Impedence spectroscopy

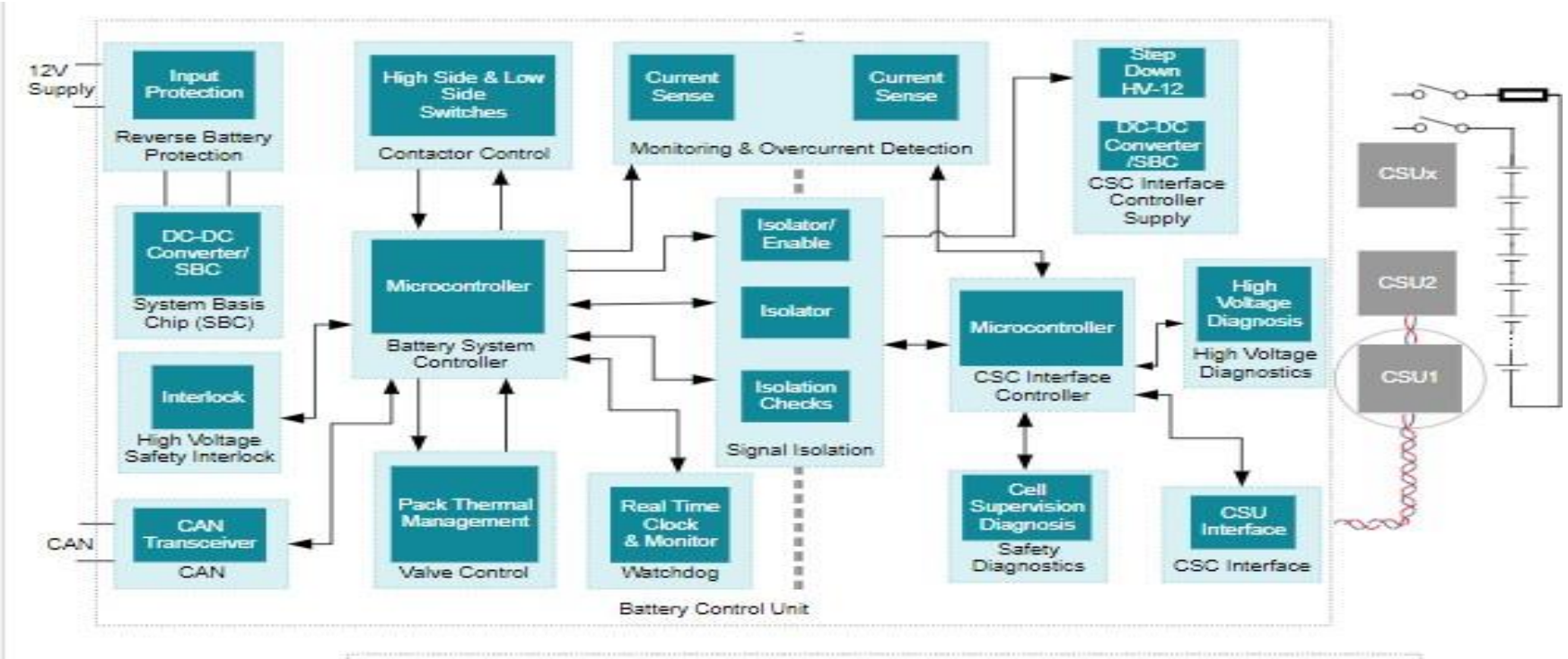
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Cell Voltage and Balancing



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High Level Functions



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Insulation Resistance

- For a DC system, a minimum insulation resistance of 100 Ω/V is
- recommended
- For an AC system, a resistance of 500 Ω/V before an insulation fault should
- be reported.
- ECE-R100 demands an insulation resistance of minimum 500 Ω/V.
- In all standards, these minimum resistances have to be measured at the
- nominal system's voltage
- It should be noted that these are the minimum required insulation resistances.
- In DIN61557-8, a minimum requirement for insulation resistance
- measurement of 15% is demanded.
- A minimum value of 1000 Ω/V and, for systems below 1000 V, a minimum insulation resistance of 1 MΩ is recommended.

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HV Isolation

- Battery packs are divided into a high voltage and a low voltage part
- Those two need to be galvanically isolated, which means digital signals need to be decoupled by either optical, inductive or capacitive means
- All thermal sensors need to be galvanically separated, in order to avoid high voltage faults to be allowed in the battery pack's low voltage sections



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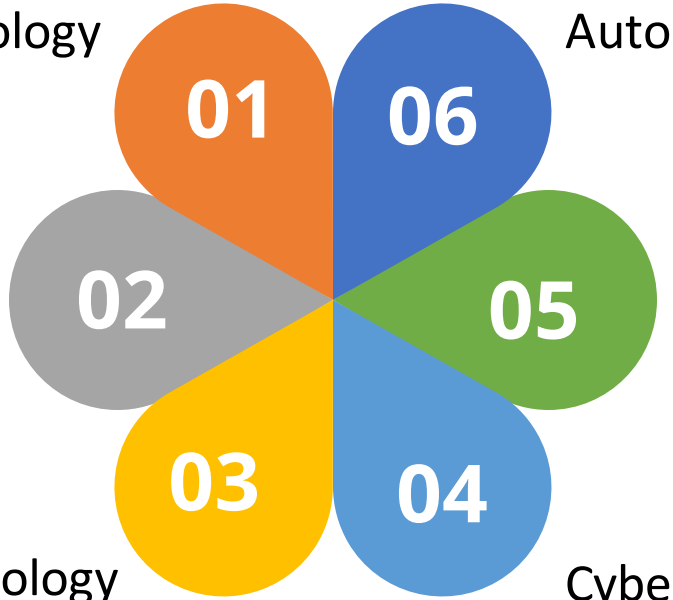
Safety Standards

No.	Standard/Guidelines	Scope and BMS Relevancy
01	ISO 6469-1:2019 Edition 3.0—Electrically propelled road vehicles—Safety specifications—Part 1: On-board rechargeable energy storage system (RESS)	This standard specifies requirements for the on-board rechargeable energy storage system for electrically propelled vehicles such as hybrid-electric vehicles, battery electric vehicles, and fuel cell vehicles. It mainly discusses climatic, electrical, functional, and simulated vehicle accident requirements. It also does not comprise motorcycles or other vehicles which are not designated as electric vehicles such as fork-lift trucks. The document does not consider the safety procedure for manufacturing, maintenance, and repair personnel.
02	ISO 6469-3:2018, Electrically propelled road vehicles—Safety specifications—Part 3: Electrical safety	This standard specifies electrical safety requirements of voltage class B electric circuit of electric propulsion systems and electrically propelled road vehicles. It specifies electrical safety requirements for protection of persons against electric shock and thermal incidents. However, it does not explain the safety measures for maintenance, manufacturing, and repair personnel.
03	ISO/TR 11955:2008, Hybrid-electric road vehicles—Guidelines for charge balance measurement	These guidelines explain the charge balance measurement procedure of hybrid electric vehicles (HEV) with batteries. The document does not consider any test for fully electric vehicles (EV). The "charge balance" term refers to the capacity of the battery.
04	SAE J2289_200807—Electric-Drive Battery Pack System: Functional Guidelines	These guidelines describe the electrical, physical, environmental, safety, and labeling requirements with product description, and shipment characteristics of the battery system for vehicles; these vehicles use a rechargeable battery to recapture the traction energy. They also explain the abnormal condition of the battery system.
05	SAE J2288_200806—Life Cycle Testing of Electric Vehicle Battery Modules	This standard describes a test method to determine the life cycle of EV battery modules. By using a set of nominal or baseline operating conditions, the expected degradation in electrical performance as a function of battery life is determined and possible failure mechanism is identified.
06	SAE J1798—Recommended Practice for Performance Rating of Electric Vehicle Battery Modules	This standard covers general test and verification methods to determine EV battery module performance. It provides the required performance in order to identify basic and minimum performance specification of EV battery modules.
07	UL 583, Standard for Electric-Battery-Powered Industrial Trucks	This standard addresses the requirements for electric-battery-powered industrial trucks, such as fork-lift trucks, tractors, platform-lift trucks, and other industrial application-specific vehicles in accordance with three most important risks: fire, explosion, and electric shock. However, there are also some other risks that are involved in this kind of equipment, such as corrosion, acid spill, and acid splash, are not discussed in the document.
08	ISO 18300:2016(E), Electrically propelled vehicles—Test specifications for lithium-ion battery systems combined with lead-acid battery or capacitor	This standard discusses the combination of lithium-ion battery with lead-acid battery or electric double-layer capacitor. The series combination of rechargeable chemical battery and double layer capacitor is an excellent solution for EV fast-charging station; however, the current and voltage spikes, during energy transfer between battery and electric double-layer capacitor, may reduce the system performance. In addition, the standard only addresses low voltage battery pack (e.g., 12V, 48V); design, test procedures, and safety requirement for high voltage battery are not covered in the document.

Source: Review of Battery Management Systems (BMS) Development ... https://www.mdpi.com › pdfx

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Our Expertise



Electric Vehicle Technology

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Future Mobility Overview

IoT, Connectivity & Advanced Computing

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